

ADR 0023 — Rehearsal workspaces

promote results, atomically

- **Status:** Proposed — direction accepted 2026-07-27; implementation scheduled after M1.11 (#64)
- **Date:** 2026-07-27
- **Milestone / issue:** post-M1.11 direction (issue to be opened when the milestone band is cut)
- **Supersedes / superseded by:** — (builds on ADR 0015/0016/0018/0019/0020/0021's write pipeline and ADR 0008's persistent clones; extends ADR 0005's structural read-only-ness from the LAN listener to the live repo itself)

Context

The operator's real workflow is not "mutate the live repo carefully" — it is "rehearse the surgery somewhere disposable, then make it real once it's right." Today git-vista offers only the careful-mutation model: every operation acts on the served repo directly, and undo reverses one operation at a time because each operation only knows how to reverse itself.

Three pressures converge:

1. **Complex operations want a rehearsal space.** A multi-step rebase/merge/ cleanup sequence should be attemptable, inspectable and discardable without the live repo ever being at risk.
2. **Undo wants to be one action.** After a rehearsed sequence goes live, "put everything back" should not require unwinding N operations in reverse order, each with its own preconditions and its own ways to fail mid-unwind.
3. **The credential story wants a smaller target.** A token that authorizes *any* write is a big secret. A token that authorizes exactly one atomic, undoable operation is a small one — and the operator wants read-only to be the default mode, with the sign-in token never printed to a terminal at all.

Decision

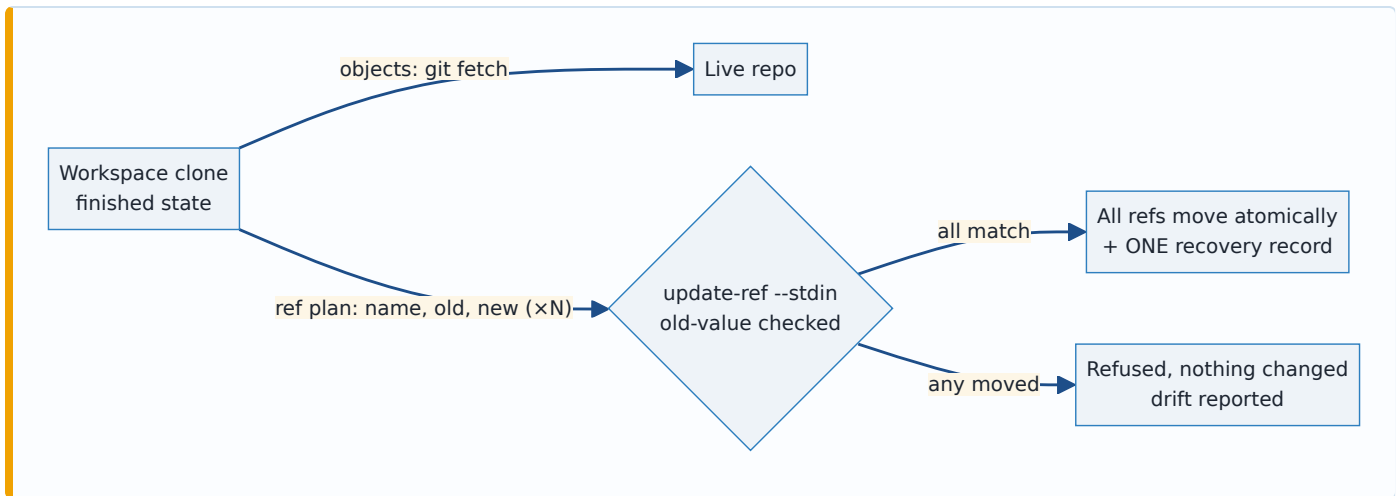
Adopt the **rehearsal workspace** model: work happens in a managed clone; the live repo is served read-only; one privileged operation — **promotion** — moves the finished state onto the live repo.

Promotion transplants the result, never replays commands

The workspace's command history is an audit record, not an execution plan. Promotion is:

1. `git fetch <workspace-path>` — the objects travel (local path remote).
2. One atomic ref transaction — `git update-ref --stdin` with **old-value checks** on every ref being moved. All refs move or none does; the transaction refuses if any ref is not where the promotion plan recorded it.

Replaying the rehearsed command list against live would mean N re-executions, each able to fail against a repo that moved, with half-applied history surgery as the failure mode. The workspace already holds the finished state; moving state is one step, moving *process* is N.



One promotion ⇒ one recovery record ⇒ undo-all

Because promotion is a single ref transaction, its recovery strategy (ADR 0021) is a single record — the pre-promotion OID of every moved ref — applied as one old-value-checked transaction in the other direction. The operator's requested "master undo-all" is therefore promotion's *native* undo, not a new mechanism layered over per-operation undo.

Drift refuses, first

If live moved since the promotion plan was built, promotion refuses — the generation gate (ADR 0018) plus the transaction's own old-value checks, which enforce it even against races inside the execution window. Presenting the drift and offering choices is future work; auto-rebasing the promotion is rejected (it reintroduces replay's partial-failure problem).

Read-only becomes structural for the live repo

The live repo's serve mode registers no write routes — the same mechanism that makes LAN sessions read-only today (ADR 0005), now applied by repo role rather than by listener. Promotion is the one privileged route, guarded by the strong credential; absent that credential the app falls closed to a fully functional read-only view. The sign-in token stops being printed to the terminal: it is handed over by the operator's auth MCP, the same 0600-file Docker-wrapper pattern already proven on his GitHub PAT, so the secret never reaches argv, environment, or scrollbar.

Alternatives considered

- **Replay the rehearsed command list onto live.** Rejected: N failure points, partial-application failure mode, and undo-all becomes an N-step unwind that can itself strand. Kept only as the audit display and as input to a future re-plan flow.
- **Auto-rebase the promotion when live drifted.** Rejected for now: same partial-failure shape as replay. Refusal is honest and cheap; choice UIs can come later.
- **Mutate live directly, harder (status quo).** Rejected as the *default* workflow by the operator's own usage: rehearse-then-promote is how he already works with clones by hand.
- **Shared object store for workspaces (`--shared` / `--reference`).** Rejected outright, not deferred: it carries a real `gc` pruning hazard against the origin, and the operator's repos are small enough (~80–100k) that a full plain clone costs nothing worth trading safety for.
- **Auto-deleting the workspace after promotion.** Rejected: the workspace is archived out of band (borg, then MEGA) by a separate MCP, and deletion happens only after that archive is confirmed. A tool that reclaims disk on its own would race that pipeline.
- **Mirroring every ref at promotion.** Rejected: it would delete live branches the workspace does not carry, which contradicts the operator's standing never-delete-branches rule. Only changed refs move.

Consequences

- The write pipeline (ADR 0015–0021) gains one operation kind: promotion. It is planned, guarded, journaled, idempotent and streamed like every other write — nothing about the pipeline changes shape.
- The M1.11 frontend operations store gains its most important customer: a promotion in flight, its outcome, and its one-touch undo are exactly the state it was built to hold.
- Live-repo write routes eventually disappear from the default serve mode — a breaking change to be sequenced deliberately, not slipped in.
- The open sub-designs were resolved with the operator the same day (recorded in the design doc): token delivery via his auth-MCP; workspaces are full plain clones (`--shared` rejected, small repos); no auto-delete — archive via his borg→MEGA pipeline, delete only on confirmation; promotion moves **changed refs only**, listed in the reviewed plan; a dirty live worktree on a moved, checked-out branch refuses promotion (undo's clean-tree guard).
- The #64 plan reserved "ADR 0023" for the frontend feature boundaries record; that record now lands as **ADR 0024** (numbers are assigned at creation time).

Where this is implemented

Not yet implemented. Direction record; the full design conversation and diagrams live in `design-docs/2026-07-27-rehearsal-workspace-promotion.md` (untracked working note). Implementation begins after M1.11 (#64) closes.

Signed: thomas2025 · 2026-07-27T01:44:04-04:00 **Updated:** thomas2025 · 2026-07-27T01:57:15-04:00 — sub-designs resolved with the operator; --shared , auto-delete and ref-mirroring moved from open to rejected.